Unit 08: Hypothesis Testing

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Example 1

Three balls will be drawn at random, with replacement, from either Basket A or Basket B; you do not know which one.

Suppose

$$H_0 : \text{The balls are from Basket A,}$$
$$H_a : \text{The balls are from Basket B.}$$

Basket A
9 red balls 1 green ball

Basket B
1 red ball 4 green balls

Reject the null hypothesis (H_0) if we observe three green balls in a row. Find the probability of a Type I error (α) and the probability of a Type II error (β).

Example 2

Purchasing shipments of iron ore pellets

μ = the mean weight of all shipments being purchased (unknown).

$\sigma = 10$ pounds (known).

The supplier claims that "the average shipment weighs 80 pounds." You suspect you may be getting shortchanged.

$$H_0 : \mu = 80 \qquad \text{versus} \qquad H_a : _____$$

Three options for H_a:

$\mu < 80$	Left-tailed or lower-tailed	Decrease ($\downarrow$)
$\mu > 80$	Right-tailed or upper-tailed	Increase ($\uparrow$)
$\mu \neq 80$	Two-tailed	Change / Different

Choose the appropriate alternative hypothesis (H_a) and perform a hypothesis test.

One-Sample Z-Test for μ

1) **Assumptions:** The population standard deviation σ is known, and the sample mean $\overline{X}$ is normally distributed or approximately normal.

2) **Hypotheses:** Choose H_0, H_a, and the significance level α.

$$H_0 : \mu = \mu_0 \qquad \text{versus} \qquad H_a.$$

Three options for H_a:

$\mu < \mu_0$	Left-tailed or lower-tailed	Decrease ($\downarrow$)
$\mu > \mu_0$	Right-tailed or upper-tailed	Increase ($\uparrow$)
$\mu \neq \mu_0$	Two-tailed	Change / Different

3) **Test statistic:**

$$Z = \frac{\overline{X} - \mu_0}{\sigma/\sqrt{n}}.$$

4) **Critical / rejection region:** α determines the total rejection-region area, and H_a determines its location.

5) **Conduct the experiment** and calculate the observed value of the test statistic.

6) **Conclusion:** If the observed test statistic falls in the rejection region, reject H_0; otherwise, fail to reject H_0.

Reject H_0: "We have sufficient evidence at significance level α to support H_a."

Fail to reject H_0: "We do not have sufficient evidence at significance level α to support H_a."

Sometimes, we also compute a confidence interval for μ after rejecting H_0 to report a range of plausible values for μ.

Example 3: Hypothesis Testing for a Proportion

A gambler has invented a new strategy for a game in which players are supposed to win 49% of the time. He hopes that his strategy will increase the probability of winning. To test it, he plays the game 80 times and wins 44 of them.
Let p denote the probability of winning a game using his strategy. Assume that the games are independent and that this probability remains constant.

(a) Choose the appropriate null and alternative hypotheses and perform a hypothesis test at $\alpha = 0.05$.

(b) If the gambler had won 55 out of 100 games, the p-value would have been _____.

Example 3 (continued): Effect of Sample Size

Comparing p-values for a fixed sample proportion $\widehat{p} = 0.55$ with different sample sizes

For the same hypotheses as in Example 3, the following table shows how the test statistic and p-value change as the sample size increases.

$$\widehat{p} = \frac{X}{n} = 0.55$$

X	n	z	p-value
44	80	1.07	0.1423
55	100	1.20	0.1151
275	500	2.68	0.0037
550	1000	3.80	≈ 0.00007

Note: The p-values are approximate and correspond to the rounded z-values shown in the table.

Example 4: Hypothesis Testing for a Variance

A manufacturer produces sticks of dynamite designed to detonate after approximately 25 seconds. For safety and consistency, the variability in detonation times should be small.
The manufacturer wants the population variance to be $\sigma^2 = 0.25$ seconds squared (equivalently, $\sigma = 0.50$ seconds), but is concerned that the variance may be larger.

Assume that the detonation times form an independent random sample from a normal population. The sample information is

$$n = 20, \qquad \overline{x} = 25 \text{ seconds}, \qquad s^2 = 0.2916 \text{ seconds}^2.$$

Choose the appropriate null and alternative hypotheses and perform a hypothesis test at $\alpha = 0.01$.